A survey about self-adaptive anomaly-detection in software-defined systems

Matthias Weiß
Institute of Industrial Automation and Software Engineering
University of Stuttgart
Stuttgart, Germany
matthias.weiss@ias.uni-stuttgart.de

Stefan Thich
Institute of Industrial Automation and Software Engineering
University of Stuttgart
Stuttgart, Germany
st173930@stud.uni-stuttgart.de

Maurice Artelt
Institute of Industrial Automation and Software Engineering
University of Stuttgart
Stuttgart, Germany
maurice-paul.artelt@ias.uni-stuttgart.de

Michael Weyrich
Institute of Industrial Automation and Software Engineering
University of Stuttgart
Stuttgart, Germany
michael.weyrich@ias.uni-stuttgart.de

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***Abstract*—The rise of software-defined automation systems is accompanied by an increase in interconnected services and networks. Among other implications, this leads to an expansion of the attack surface for cyber threats, which in turn can lead to significant security issues and financial losses. Against this background, this paper investigates the effectiveness of anomaly-based intrusion detection systems (IDS) for detecting and preventing new and evolving cyber threats. The paper especially considers the challenge of concept drift, which describes the phenomenon that the statistical properties of the network traffic or behavior patterns that an IDS is trying to detect change over time. To accomplish this, a comprehensive literature review, prioritizing recent studies since 2019, highlights the challenges of concept drift and the application of reinforcement learning in IDS. The results suggest that the integration of adaptive strategies and reinforcement learning in IDS enables significant improvements in anomaly detection.**

Keywords—self-adaptive anomaly detection, reinforcement learning, intrusion detection, concept drift

I. Introduction

In the landscape of modern and future automation systems, the proliferation of connected IT infrastructures leads to an increased amount of network interfaces, which inevitably expands the attack surface for cyber threats [1]. This expansion is not without significant financial implications; statistical data estimates that between 2019 and 2023, the global economy lost around $5.2 trillion in revenue due to cyberattacks [2]. Such figures prompt a critical examination of the development of more robust defenses against emerging cyber threats. One approach to minimize the problem are anomaly-based intrusion detection systems (IDS), which serve as a central mechanism for identifying anomalies and mitigating unauthorized intrusions [3]. However, the effectiveness of these systems is continually challenged by the dynamic nature of cyber threats, particularly the phenomenon known as concept drift, where the behavior of network traffic evolves over time, potentially rendering existing detection models obsolete [4].

This introduces the central research question of this paper: *How can the effectiveness of anomaly-based intrusion detection systems (IDS) be improved in the context of increasingly connected driving to detect and prevent new and evolving cyber threats, while accounting for concept drift and minimizing false positives?*

There are several challenges in addressing this issue. Anomaly-based IDSs are at the forefront of detecting novel attacks that have not been previously identified or catalogued. Unlike misuse based IDSs, which rely on known patterns of malicious behavior, anomaly-based systems must discriminate between benign and malicious activity in an environment where data characteristics and attack methods can change rapidly [5]. Recent approaches in IDS research have focused on overcoming these challenges. This paper aims to provide an overview of different techniques to improve the system to better cope with the concept mismatch phenomenon and to reduce the false positive rate.

The rest of the paper is structured as follows: Chapter 2 presents the methodology used to select the papers presented and explained in Chapter 3. Chapter 4 discusses the results and outlines an approach to self-adaptive anomaly detection that addresses the previously found research gaps. The paper is summarized and an outlook is given in chapter 5.

II. Methodology

For this paper, a systematic literature review according to the Kitchenham methodology [6] is applied to analyze recent advancements and challenges in self-adaptive anomaly detection. The review was conducted using the IEEE Xplore Digital Library, covering the period from 2019 to 2024 to capture both current findings and technological advancements.

In the context of the paper's research question mentioned in chapter 1, two critical aspects emerge from the initial pre-screening of publications as central to advancing the field of anomaly detection in IDS systems: the challenges of concept drift and the reduction of false positives. As such, two separate search procedures were performed, which are described in the following.

A. Concept Drift in Anomaly Detection

Concept drift represents a fundamental challenge in the realm of anomaly detection, where the underlying data distribution changes over time [7]. This phenomenon can severely impact the performance of predictive models by making previously learned patterns obsolete [8, 9]. To counteract this, there is a pressing need for adaptive strategies that can recognize and adjust to concept drift effectively.

Keywords: The keywords used for the search were "anomaly detection", "adaptive" and "concept drift". These terms were chosen to identify papers that specifically address adaptive anomaly detection, considering changing data patterns.

Results: The initial search returned a total of 41 hits. To increase the relevance of the results, a filter was first applied

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to the titles of the papers, which reduced the number to 20 relevant publications. The abstracts of these 20 papers were then read to assess their suitability to the objectives of this search. Based on this assessment, 4 papers were identified as particularly relevant and selected for detailed analysis.

B. Reducing False Positives in Anomaly Detection

The challenge of improving the effectiveness of anomaly-based intrusion detection systems (IDS) lies primarily in reducing false positives (FP) [10]. These FPs are not only annoying but can also lead to a significant waste of resources as security teams spend time and energy investigating events that pose no real threat. A promising approach to this challenge is the application of reinforcement learning (RL), which imposes stricter penalties for generating FPs, thereby increasing the accuracy of the IDS. As such, the

Keywords: The search used specific terms such as "reinforcement learning", "adaptive" and "anomaly detection". This selection aimed to identify studies that focus on improving anomaly detection systems through adaptive methods and reinforcement learning.

Results: The initial search returned 35 hits. A targeted filter based on the titles narrowed this down to 15 relevant articles. Detailed analysis of the abstracts of these articles led to the selection of three publications that presented innovative approaches to minimizing FPs through reinforcement learning. These works were chosen for their direct relevance to improving precision in anomaly-based reinforcement learning. Among the explored literature, a standout paper and published in 2022. The focus was on penalizing false positives and false negatives, and the methods and results presented met our requirements.

III. Survey Results

Following the methodology of the systematic literature review, this chapter presents the results of our review. For both the challenge of concept drift and the reduction of false positives, two papers are presented thoroughly, followed by a list of alternative approaches. The beginning of each section presents the preliminary steps used in the presented approaches, followed by the scheme that is actually proposed.

A. Concept Drift in Anomaly Detection

Das and Pasikhani present an approach to adaptive Intrusion Detection in Acoustic Sensor Networks [11]. Technologies used in their system include the adaptive random forest (ARF), the adaptive windowing algorithm (ADWIN), and the k-dominant query tree (kdqTree), which are described in the following.

ADWIN: ADWIN is an algorithm that dynamically adjusts the size of a window (W) that grows as long as the data pattern remains unchanged and shrinks when a change is detected. Based on the data distribution, the algorithm tries to find two sub-windows within W (w0 and w1) that have different averages.

kdqTree: A kdqTree, short for k-dominant query tree, is a specialized data structure designed to efficiently manage and query multi-dimensional data [12]. It extends the principles of k-d trees (k-dimensional trees) by focusing on queries where certain dimensions are given more importance or dominance, over others. This makes kdqTrees particularly useful for dealing with datasets where not all attributes are equally relevant to every query. In the context of concept drift - a scenario where the statistical properties of the target variable that the model is trying to predict change over time - a KDQ tree proves to be very promising [9]. Concept drift can significantly impact the performance of predictive models by rendering previously learned data patterns obsolete or less accurate. The kdqTree helps to overcome this challenge by allowing more efficient data organization and retrieval based on the current 'dominant' attributes or features of the data [11]. This capability ensures that the model can adapt more quickly and efficiently to changes in data patterns, maintaining its accuracy and relevance in dynamic environments. By focusing on the most relevant dimensions of the data, a KDQ tree helps to identify and respond to concept drift, making it an essential tool for adaptive learning systems that need to operate effectively in changing conditions.

ARF: Adaptive Random Forest Classifier is a machine learning model that combines several Hoeffding trees. When making a final decision, it gathers opinions from all these trees and selects the result with most votes as the result. This model is particularly adept at adapting to changes or "drifts" in the data over time. It incorporates a special mechanism for detecting these changes, known as the ADWIN algorithm, to remain relevant and accurate. ADWIN is connected to the ARF. When ADWIN detects a potential change, it alerts the model and causes new Hoeffding trees to be generated in the background. If the change is confirmed to be significant, these new trees replace the current ones, ensuring that the model remains effective. In addition, the model is designed to remember previous configurations so that it can adapt quickly if past conditions recur. It also uses strategies such as resampling and selecting different features for splitting to maintain its performance in diverse or unbalanced data sets.

Proposed Scheme: Figure 1 shows a technical illustration of the model. The One-Class Support Vector Machine (OCSVM) anomaly detector first filters the incoming data to allow normal behavior to pass through and be included in the final prediction. Anomalous data is then analyzed by an ARF classifier, which uses an ensemble of Hoeffding trees together with a drift detection mechanism - ADWIN and kdqTree. If there is no drift, the ARF predictions are used for the final decision. If drift is detected, ADWIN adjusts the ARF by updating its tree ensemble and incorporating these changes into the final prediction. In a second phase of anomaly detection, a bagging ensemble of OCSVMs is used to identify outliers by refining the decision boundary to minimize false positives. The final decision process integrates the initial OCSVM outputs for normal instances, ARF predictions and ensemble OCSVM analysis to classify the data as either normal or indicative of an attack. If an attack is identified, the system alerts a cryptographically secure intrusion prevention system (IPS) for further action.

B. Reducing false positives in anomaly detection

Zhang *et al.* [5] perform anomaly detection in time series by selecting models via a reinforcement learning-based classifier. This section focuses first on the unsupervised detection of anomalies in time series. Subsequently, the area of reinforcement learning (RL) is explored, using the mathematical framework of a Markov decision processes.

Unsupervised Anomaly Detection in Time Series: Time series data, represented as a sequence X consisting of either single values or vectors at each time point, are analysed for

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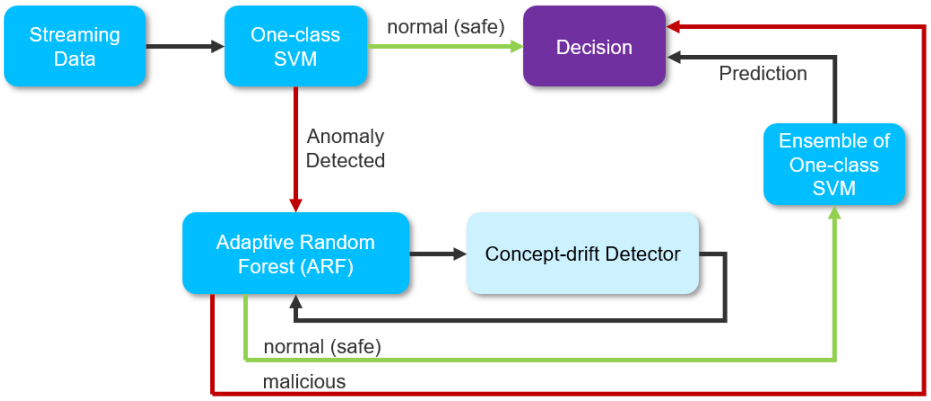


Fig. 1. Concept Drift Anomaly Detection based on [11]

anomalies using a particular approach. This method involves first training a model on a series of data known to be normal, and then applying the model to a new set of data that may contain abnormal instances. The aim during training is to familiarize the model with what constitutes normal behavior. When tested on a potentially contaminated dataset, the model generates an anomaly score for each data point. By setting a pre-defined threshold, it becomes possible to distinguish the normal from the anomalous, thereby categorizing each instance based on its score.

Reinforcement Learning and Markov Decision Process: Reinforcement learning (RL) is a machine learning approach that focuses on solving sequential decision problems. Its goal is to enable an agent to find the best actions within an environment to maximize cumulative rewards, typically formulated by a Markov Decision Process (MDP). An MDP is characterized by a set of states (S), actions (A), state transition probabilities (P), rewards (R), and a discount factor (γ) that influences the calculation of future rewards and typically ranges between 0 and 1. A policy, π(a|s), guides the agent by mapping current states to probabilities of taking certain actions, with the goal of maximizing the expected total reward.

Proposed Scheme: The model selection problem was formulated as an MDP. Figure 2 shows the MDP formulation and the definitions of state, reward and action are presented in the following section.

State: The state space in this context is equivalent in size to the model pool, with each state representing a selected anomaly detector. The state variables include a variety of factors, including the scaled anomaly score and threshold (both normalized to the [0, 1] range using min-max scaling), the binary prediction label (where 0 indicates normal and 1 indicates anomalous), the confidence in the distance to the threshold, and the confidence in the prediction consensus.

Reward: The reward function is defined by comparing the predicted label to the actual label and is represented by four values (true positive, true negative, false negative, false positive). With the reward function, it is possible steer the model selection in a certain direction to reduce the FP rate, which is a problem in anomaly-based IDS systems.

Action: The action space is discrete and corresponds to the size of the model pool. In this setup, each action involves selecting a candidate detector from the pool, identified by the index of the selected model within the pool.

C. *Alternate Methods*

LSTM Mitigator: This approach uses Long Short-Term Memory (LSTM) layers. These are described in more detail in [13]. The paper describes that LSTM layers can detect long-term dependencies in the data and belong to the category of deep learning, which makes them very efficient for large data sets. A method has been developed that offers a powerful alternative to conventional approaches to time series analysis. However, it also has drawbacks. For example, LSTMs are very complex and computationally intensive, so they cannot be used in small systems with low local computing power. I see their application more in stationary areas with a lot of computing power and a high throughput of sensors, for example in an autonomous factory floor.

SALAD: Self-Adaptive Lightweight Anomaly Detection (SALAD), as described in [14], is used to transform complex time series into average absolute relative error (AARE) series. For each new data point in the time series, the absolute relative error is calculated against a predicted value. The ARE of a data point is typically expressed as the ratio of the absolute error to the actual value, which allows the error to be normalized to make it comparable regardless of the size of the data. The ARE is then calculated as the average of the ARE values over the most recently observed data points. This averaging helps to compensate for short-term fluctuations and focuses on the general trend of the error over time. The self-adaptive detection threshold dynamically adapts to the observed ARE values. This allows the system to automatically calibrate its sensitivity to detect anomalies based on the current data. The disadvantage of this method is that the use case was designed for time series, which is appropriate from an approach point of view as it reduces computational complexity, but it is difficult to represent network logs and attacks as a time series.

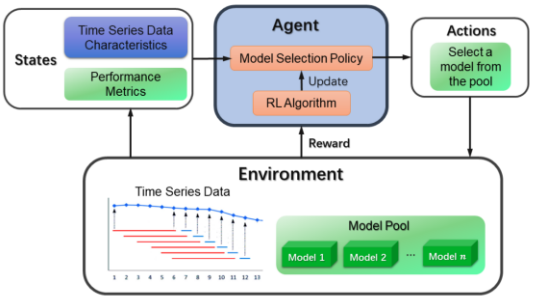


Fig. 2. Model Selection of Anomaly Detectors MDP [5]

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IV. Research Gaps

A. Concept Drift Problem

Once the technologies have been presented, we evaluate how well they meet our requirements. Adapting this technology to the automotive sector is challenging. In contrast to maritime applications where acoustics play a dominant role, automotive sensor technologies are more diverse and must deal with a wider range of environmental conditions and sources of interference. Regarding the use of data sets, it is important to note that synthetic or simulated data sets are often used in initial research and development to test certain technologies. The UW-ASN dataset used in this context, which is not a real-world dataset, was probably used to evaluate the functionality and efficiency of the underwater acoustic sensor IDS under controlled conditions. While such datasets provide valuable insights into the performance and potential weaknesses of systems, benchmarking performance against real-world datasets is critical to validate operational suitability.

B. False Positive Problem:

As effective as standard models such as OCSVM or iForest may be, the complexity and dynamics of modern cyber threats require more advanced and adaptive approaches. One interesting advance in this area is the integration of "reinforcement pool" approaches, which include concepts such as the Concept Drift Detector. Another innovative step is the addition of an adversarial and deep learning component. Adversarial approaches utilize techniques developed in Generative Adversarial Network (GAN) research to train models that are more resistant to manipulative attacks. This methodology simulates attack and defense strategies during the training process, allowing the system to learn to detect even the most sophisticated intrusion attempts. Deep learning, in particular Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), complement traditional IDS models with their ability to recognize complex relationships in large amounts of data.

C. Proposed Future Direction

To close the highlighted research gaps, the authors suggest combining methods of concept drift detection and reinforcement learning. The purpose of the latter is specifically to minimize the false positive rate, a known problem with anomaly-based IDS systems. The reinforcement learning pool utilizes technologies such as the concept drift detector presented in chapter 3, which is particularly strong at detecting and eliminating concept drift, as well as other technologies. The false positive rate is penalized more severely to improve the accuracy and efficiency of the system in detecting real threats while minimizing the burden of false alarms.

V. Conclusion and Outlook

In this paper, a literature review was conducted focusing on self-adaptive anomaly detection systems in software-defined automation systems, specifically intrusion detection systems.

The areas of concept drift and reinforcement learning were investigated, as concept drift and its changes play an important role in adaptivity, and intrusion detection systems have a high positive error rate that can be regulated with rewards. Alternative approaches were also investigated, such as an LSTM mitigator method for anomaly detection, and a method where complex time series can be transformed into AARE series, reducing computational intensity. Based on the literature review, current research gaps in the IDS field were identified. To close them, the methods are combined into one concept to highlight possible directions for future research. In this context, the authors outline a system that allows concept drift detection, regulates the false positive rate with reinforcement learning or adaptive thresholds, and adds an adversarial and deep learning component.

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